Course Name:	**Database Management System Laboratory**	**Semester:**	**IV**
Date of Performance:	**03/02/2026**	**Batch No:**	**B2**
Faculty Name:	**Prof. Megha Sharma**	**Roll No:**	**16010124 107**
Faculty Sign & Date:		**Grade/Marks:**	**25/25**

Experiment No: 1

Title: Entity-Relationship diagram

Aim and Objective of the Experiment:
Aim: Problem Definition and Design of Entity-Relationship diagram **Objective:** To define a Database Problem and Design an ER diagram for a business domain.

COs to be achieved:
CO1 : Design entity-relationship diagrams to represent different database application scenarios.

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Books/Journals/Websites referred:
1. G. K. Gupta :"*Database Management Systems*", McGraw – Hill 2. Korth, Slberchatz, Sudarshan : "Database Systems Concept", 6th Edition , McGraw Hill 3. Elmasri and Navathe, "Fundamentals *of Database Systems*", 5thEdition, PEARSON Education.

Tools required:
Dia Software: A software to Design ER Model Dia is one of the convenient open source tool which runs on multiple platforms including Linux, Windows and MacOS.Dia has a number of "sheets" each of which includes diagram

objects for different modeling tools, such as UML, ER diagrams, flowcharts, etc.

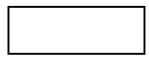
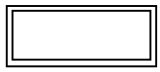
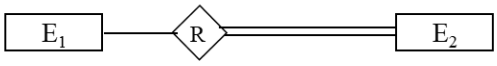
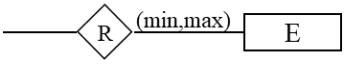
Theory:

The ER tool has objects for entities, relationships, attributes (using the oval notation), edges, and so on. The properties boxes for each of these elements allow you to specify cardinality constraints, total participation, identifying relationship, etc.

It supports many common formats to store diagrams such as jpeg, png, eps, etc.

Pre Lab/ Prior Concepts:

The ER data model was developed to facilitate the database design by allowing specification of an enterprise schema that represents the overall logical structure of the database. The ER model is one of the several data models. The semantic aspect of the model lies in its representation of the meaning of the data. The ER model is very useful many database design tools drawn on concepts from the ER model. The ER model employs 3 basic notations: entity set, relationship set and attributes.

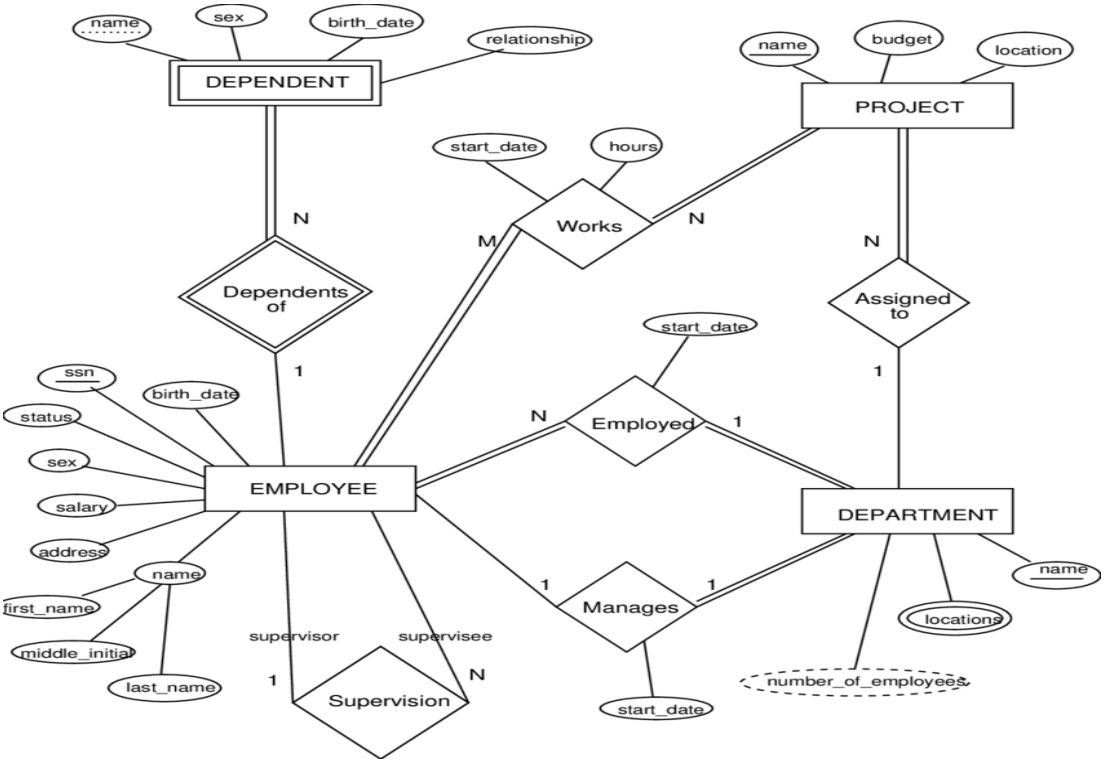
Symbol	Meaning
	ENTITY TYPE
	WEAK ENTITY TYPE
	RELATIONSHIP TYPE
	IDENTIFYING RELATIONSHIP TYPE
	ATTRIBUTE
	KEY ATTRIBUTE
	MULTIVALUED ATTRIBUTE
	COMPOSITE ATTRIBUTE
	DERIVED ATTRIBUTE
	TOTAL PARTICIPATION OF E_2 IN R
	CARDINALITY RATIO 1:N FOR E_1:E_2 IN R
	STRUCTURAL CONSTRAINT (min, max) ON PARTICIPATION OF E IN R

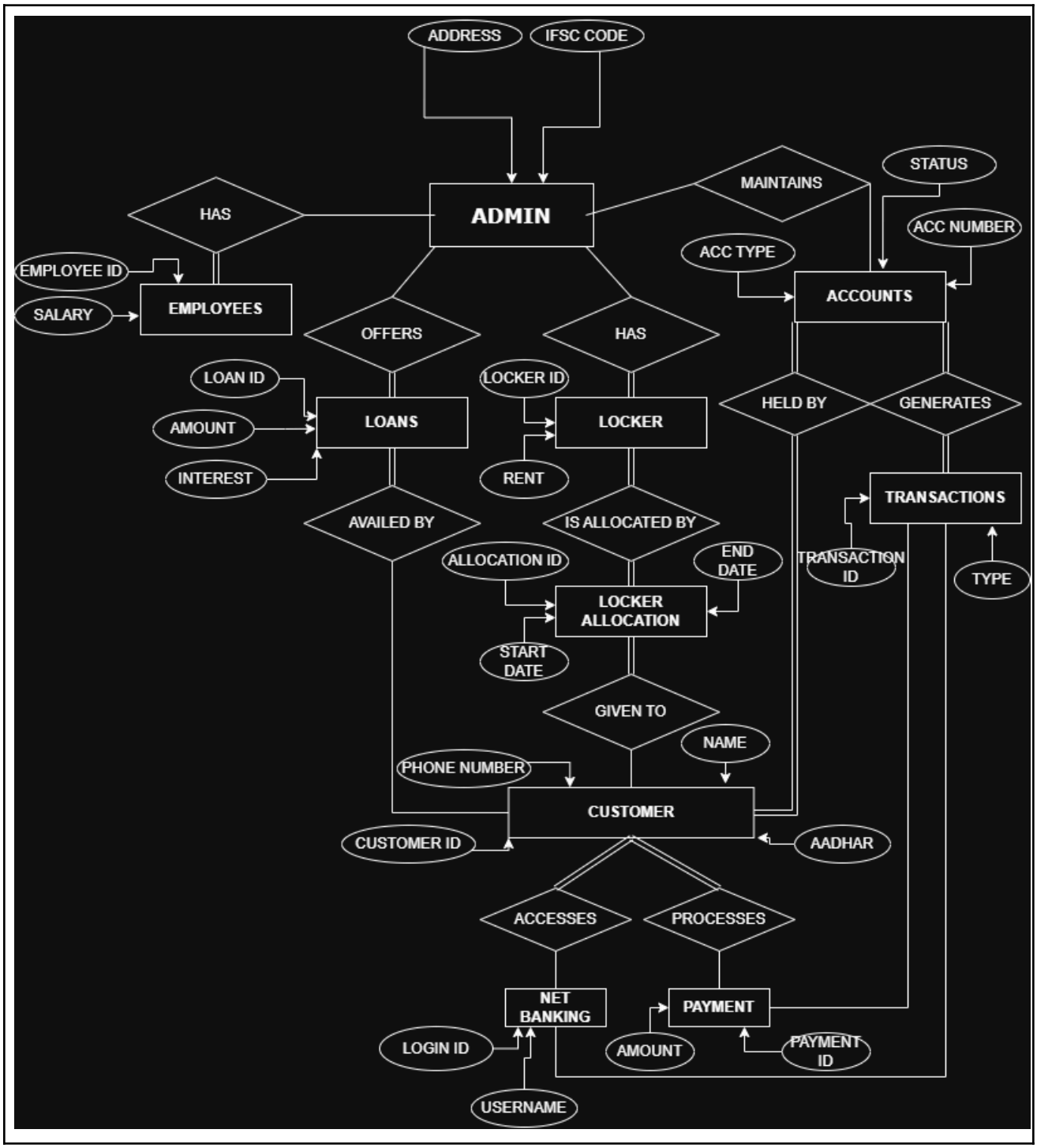
Implementation details:

Procedure for doing the ER diagram experiment

1. Identifying the Entities (Strong and weak entities)
2. Identify attributes of the Entity (keys, partial key, simple, composite, multivalued, derived)
3. Identify relationship(recursive)
4. Identify the structural constraints of the relationship (cardinality ratio, participation constraints)

ER- Diagram for company Case Study Database:

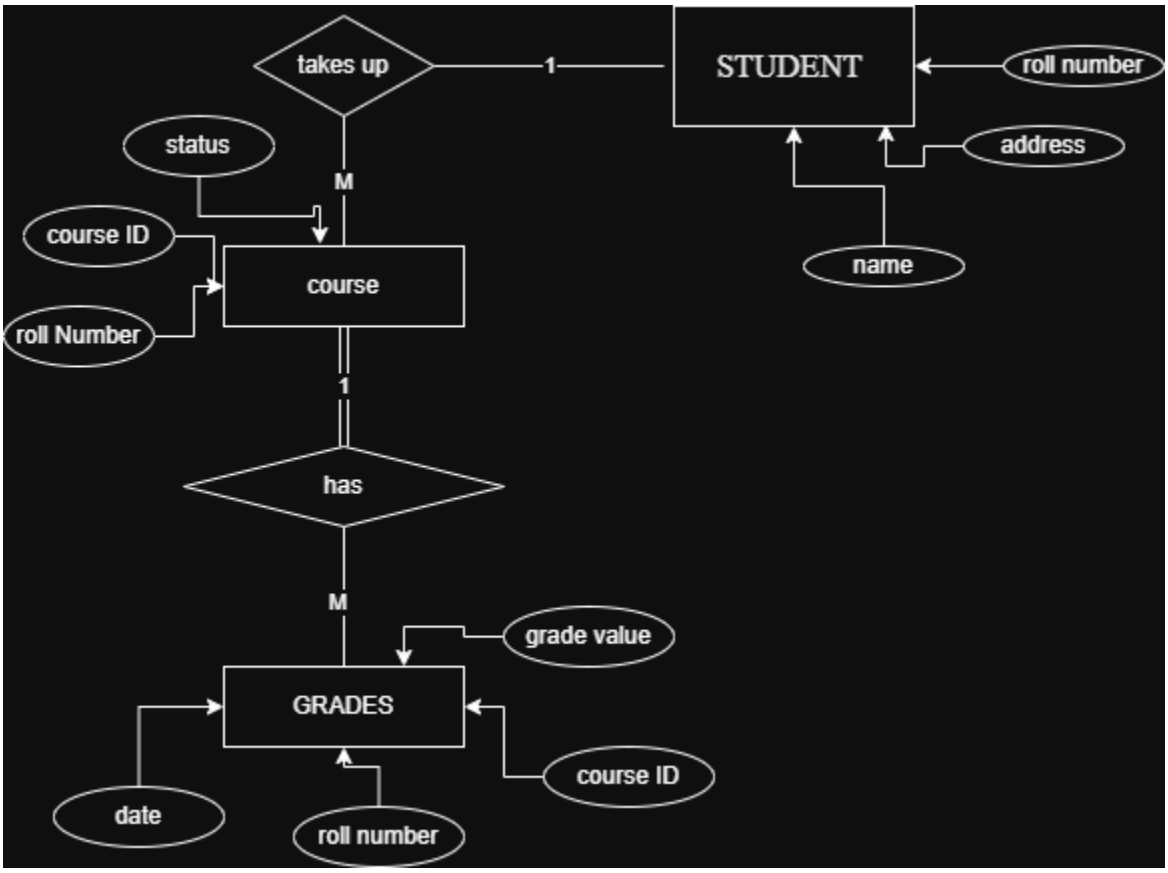



ADDRESS
IFSC CODE
ADMIN
HAS
MAINTAINS
STATUS
ACC NUMBER
ACC TYPE
ACCOUNTS
EMPLOYEE ID
SALARY
EMPLOYEES
OFFERS
HAS
LOAN ID
AMOUNT
INTEREST
LOANS
LOCKER ID
LOCKER
RENT
HELD BY
GENERATES
TRANSACTIONS
TRANSACTION ID
TYPE
AVAILED BY
IS ALLOCATED BY
ALLOCATION ID
END DATE
LOCKER ALLOCATION
START DATE
GIVEN TO
NAME
PHONE NUMBER
CUSTOMER
CUSTOMER ID
AADHAR
ACCESSES
PROCESSES
NET BANKING
PAYMENT
LOGIN ID
AMOUNT
PAYMENT ID
USERNAME

Post Lab Subjective/Objective type Questions:

Post Lab Descriptive Questions (Add questions from examination point view)

1. In the Academic database a Grade is issued to each STUDENT for each COURSE taken and stored in the STUDENT COURSE DETAIL entity. A STUDENT may decide to re-take a COURSE to better their GRADE. The administration would like to keep a record of the old/previous Grade as well as the new Grade. Show ER diagrams to include historical Grades if the students should have them.



2. Discuss the concept of aggregation. Give an example. How to represent aggregation in the ER model.
   Aggregation is an abstraction through which we can represent relationships as higher-level entity sets.
   Example: an Employee working on a project may require some machinery. So, REQUIRE relationship is needed between the relationship WORKS_FOR and entity MACHINERY. Using aggregation, WORKS_FOR relationship with its entities EMPLOYEE and PROJECT

is aggregated into a single entity and relationship REQUIRE is created between the aggregated entity and MACHINERY.

To represent aggregation in the ER model,

1. Create Schema for the Aggregated Relationship
2. Create Schema for the Higher-Level Relationship (Aggregation)

3. Two separate banks which decide to merge. Both banks use the same ER database schema(Assume the ER diagram). If the merged bank is to have a single database, there are several potential problems:
   - The possibility that two original banks have branches with the same name
   - The possibility that some customers are customers of both original banks
   - The possibility that some loan or account numbers were used at both original banks

   Discuss for each of these potential problems , why there is indeed potential difficulty in the database based on the ER model. Propose a solution to a problem. For your solution, explain any changes that would have to be made and describe what their effect would be on the ER database schema and the data.

1. Allocate the Primary key of different branches to be an IFSC code or a branch ID to differentiate.
2. Make customer ID the foreign key for the account numbers, allowing the customers to have multiple accounts. Then, all bank related entities could be related to the account number instead of the customer ID treating the same person as two accounts
3. Account numbers are now linked with the IFSC codes to avoid overlapping. Loans also have a branch ID foreign key to differentiate between branches

Singular solution: Could have a separate value at a higher segregation level with primary key Original_Bank. All other conflicting entities take up this one as a foreign key to differentiate.

Conclusion:

The experiment successfully implemented ER diagram building and schema relationship management.